

Problem

The Laplace transform of $u(t - 2)$ is:

(a) $\frac{1}{s + 2}$

(b) $\frac{1}{s - 2}$

(c) $\frac{e^{2s}}{s}$

(d) $\frac{e^{-2s}}{s}$

Step-by-step solution

Step 1 of 1

$$(d) \quad L[4(t-2)] = \int_0^{\infty} 4(t-2)e^{-st} dt$$

$$\begin{aligned} 4(t-2) &= 0 & \text{for } t < 2 \\ &= 1 & \text{for } t > 2 \end{aligned}$$

$$L[4(t-2)] = \int_0^{\infty} e^{-st} dt = \frac{-1}{s} e^{-st} \Big|_2^{\infty}$$

$$= \frac{e^{-2s}}{s}$$